University of Florida, Department of Mathematics

Algebra First-Year Exam, May 2023, Part 1

Answer four problems. If you turn in more than four, only the first four will be graded. Within reason, you may use theorems as long as you state them clearly.

1. Let $P \subset S_7$ be a Sylow 7-subgroup. Show that the normalizer N of P is a semidirect product $\mathbb{Z}/7\mathbb{Z} \rtimes \mathbb{Z}/6\mathbb{Z}$, where the homomorphism $\mathbb{Z}/6\mathbb{Z} \to \mathrm{Aut}(\mathbb{Z}/7\mathbb{Z})$ is injective.

2. It is known that A_7 has a simple subgroup of order 168. Show that this subgroup is maximal.

3. **Warning: Suspected error.** The source clearly states that one should deduce that every group of order 315 is abelian, but this conclusion appears mathematically false; nonabelian groups of order 315 exist. The intended statement is not uniquely determined, so the source reading has been retained.

Suppose G is a group of order 315.

(i) Show that if G has a normal Sylow 3-subgroup, that subgroup is contained in the center of G.

(ii) Deduce from this that G is abelian.

4. Suppose $\sigma \in A_8$ has order 15.

(i) Find the order of the centralizer of σ.

(ii) Find the number of conjugates of σ.

5. Suppose A is an abelian group of order p^4, where p is a prime.

(i) Show that A has at most $1 + p + p^2 + p^3$ subgroups of order p.

(ii) Show that equality happens if and only if every nonidentity element of A has order p.

6. Suppose G is a group and M, N are normal subgroups of G such that $G = MN$. Show that $G/(M \cap N) \simeq G/M \times G/N$.